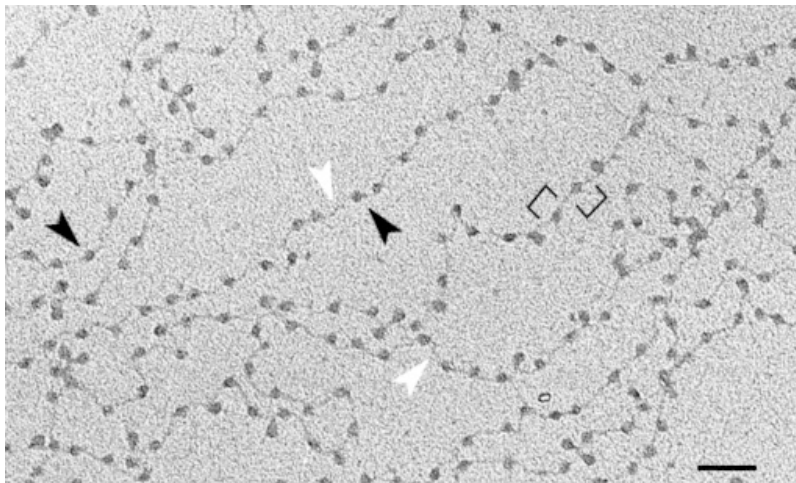

Question #: 1

A researcher is studying the structure of chromatin using electron microscopy. An example of one of these images is shown which illustrates how DNA wraps around a group of proteins to form a nucleosome unit. The amino acid composition of these proteins helps in their physiological function. Which of the following amino acid residues most likely facilitates this interaction?

- A. Branched chain amino acids like leucine
- B. Negatively charged amino acids like aspartate
- ✓C. Positively charged amino acids like lysine
- D. Thiol group containing amino acids like cysteine
- E. Hydroxyl group containing amino acids like serine
- F. Aromatic amino acids like phenylalanine

Rationale: Histones are rich in the positively charged amino acids like lysine and Arginine which facilitate the packaging of negatively charged DNA around them. Acetylation of the lysine residues in Histones reduces the binding of DNA to the histones

Attachment:



53603 ExamSoft image.JPG

Question #: 2

An investigator is analyzing the changes in a ciprofloxacin-resistant strain of *Pseudomonas aeruginosa*. The DNA gyrase of the bacteria is purified. It is shown to have a variant amino acid sequence when compared to the enzyme in the ciprofloxacin-sensitive strain. The drug no longer interacts with the variant DNA gyrase. Which of the following is the most likely function of the enzyme in the ciprofloxacin-sensitive strain?

- A. Cleaves single-stranded DNA
- ✓B. Cleaves double-stranded DNA
- C. Facilitates formation of heterochromatin
- D. Catalyzes the formation of the peptide bond
- E. Causes termination of RNA synthesis

Rationale: DNA gyrase is an example of DNA topoisomerase-II. This enzyme cleaves ds-DNA and reduces supercoiling.

DNA topoisomerase-I cleaves one of the strands of DNA and reduces supercoiling.

Heterochromatin formation occurs when DNA is tightly wound around histones. Heterochromatin formation occurs when histones are deacetylated.

Peptide bond formation during translation is catalyzed by peptidyl transferase activity on the larger subunit of the ribosome. Inhibited by chloramphenicol in prokaryote translation. Cycloheximide inhibits eukaryotic peptidyl transferase activity.

RNA synthesis in bacteria is catalyzed by RNA polymerase. Prokaryotic RNA polymerase inhibitor is Rifamycin (or Rifampin).

Question #: 3

An investigator isolates a bacterial strain that is found to have reduced rates of mRNA synthesis. Which of the following proteins is most likely to exhibit reduced function in this strain?

- A. Dna A protein
- ✓B. Sigma factor
- C. Single strand binding proteins (SSB)
- D. Initiation factor-1 (IF-1)
- E. Peptidyl transferase
- F. Elongation factor-G

Rationale: Sigma factor is the only protein in the choices provided which is required for transcription. Sigma factor binds to the promoter sequence of the gene and helps in unwinding the DNA. Dna A protein and SSB are required for replication, whereas IF-1, EF-G and peptidyl transferase are required for translation

Question #: 4

A pharmaceutical company develops a drug that inhibits the enzyme which catalyzes the formation of a phosphodiester bond between a 3'-OH group of a nucleotide and a nucleoside monophosphate of a polynucleotide with ATP hydrolysis. Which of the following enzymes is most likely inhibited by this drug?

- A. DNA polymerase I
- ✓B. DNA ligase
- C. Telomerase
- D. RNA polymerase
- E. Helicase
- F. Peptidyl transferase
- G. Aminoacyl tRNA synthetase

Rationale: DNA ligase catalyzes the formation of a phosphodiester bond - this is important in the synthesis of the lagging strand to join the neighboring Okazaki fragments. DNA Polymerase-I catalyzes the formation of a phosphodiester bond between 3'-OH group of a nucleotide and a dNTP with hydrolysis of pyrophosphate

Question #: 5

An enzyme with phenylalanine at amino acid position 206 is isolated. A variant enzyme is identified in the population in which a change in amino acid sequence at this position is observed, but there is no change in the charge and electrophoretic properties of the enzyme. Which of the following amino acids is most likely replaced in the variant enzyme?

- A. Lysine
- B. Glutamate
- C. Aspartate
- ✓D. Tryptophan
- E. Arginine
- F. Histidine

Rationale: Phenylalanine is an uncharged amino acid. Tryptophan is the only amino acid in the list that is also uncharged.

Lysine, Arginine and Histidine are positively charged amino acids.

Glutamate and aspartate are negatively charged amino acids.

Question #: 6

A 15-day-old boy is brought to the emergency room by his parents because of a 30-minute history of seizures. Physical examination shows a lethargic infant. Laboratory studies show the presence of branched chain keto acids in urine. A special formula with restricted amounts of branched chain amino acids is prescribed. Dietary restriction of which of the following amino acids is most likely advised?

- A. Serine and glycine
- B. Glutamate and aspartate
- ✓C. Leucine and valine
- D. Methionine and cysteine
- E. Lysine and leucine
- F. Cysteine and phenylalanine
- G. Arginine and histidine

Rationale: Leucine, valine and isoleucine have R-groups that are uncharged and are branched chain amino acids. Serine contains a hydroxyl group; Glutamic acid is a negatively charged acidic amino acid; Lysine and arginine are positively charged and basic amino acids; Methionine and cysteine are sulfur containing amino acids.

Question #: 7

A 7-year-old girl is diagnosed with type I diabetes mellitus. Genetic testing shows a point mutation, leading to the substitution of a highly conserved cysteine residue at amino acid position 7 in the B-chain of insulin. Which of the following is most likely affected as a result of this mutation?

- ✓A. Formation of a disulfide bridge critical for folding of proinsulin
- B. Formation of hydrogen bonds between the A chain and B chain
- C. Formation of preproinsulin by the ribosome
- D. Removal of the amino-terminal signal sequence of preproinsulin
- E. Delivery of proinsulin to the endoplasmic reticulum
- F. Splicing of the insulin m-RNA

Rationale: Cysteine in peptides is useful for the formation of inter and intra chain disulfide bonds. The A and B chains of insulin are linked by disulfide bonds.

Question #: 8

An investigator is analyzing tissue specific processing of the alpha-tropomyosin pre-mRNA which results in different mRNAs containing different combinations of introns and exons. It is observed that the alpha-tropomyosin protein in striated muscle is different compared to the alpha-tropomyosin protein in smooth muscle. Which of the following processes is most likely responsible for this difference?

- A. RNA editing
- ✓B. Post-transcriptional modification
- C. Post-translational modification
- D. Proteolytic cleavage
- E. Epigenetic modification

Rationale: Alternative splicing (post transcriptional modification) is responsible for the formation of various types of alpha tropomyosin. Do not mix-up post-transcriptional processing (modifications of RNA) and post-translational modifications (Protein modifications)

Question #: 9

A 2-year-old girl is brought to the physician by her parents because of a 1-month history of episodes of lethargy that occur approximately 4 hours following consumption of fruits. She is diagnosed with an inherited disorder of fructose metabolism. Sequencing of the mRNA coding for an enzyme required for fructose metabolism shows a single point mutation in a codon (AGU? AGA). Which of the following amino acids is most likely present at this position in the altered protein in the child?

- A. A negatively charged amino acid
- B. A sulfur containing amino acid
- C. A hydroxyl group containing amino acid
- ✓D. A positively charged amino acid
- E. An aromatic amino acid
- F. A branched-chain amino acid
- G. A hydrophobic amino acid
- H. An acidic amino acid

Rationale: AGA codes for Arginine (use the genetic code provided). Arginine is a positively charged amino acid. Review the amino acids belonging to the other classes

Negatively charged (acidic) amino acids: Glutamate and aspartate

Sulfur containing amino acids (uncharged): Methionine and cysteine (thiol group)

Hydroxyl group containing amino acids (uncharged, hydrophilic): Serine, threonine and tyrosine (aromatic and hydroxyl group side chain)

Positively charged (basic) amino acids: Histidine, Lysine and Arginine

Aromatic amino acids: Phenylalanine (hydrophobic uncharged), Tyrosine (Aromatic and contains hydroxyl group) and Tryptophan (hydrophobic uncharged)

Branched chain amino acids (Unchanged and hydrophobic): Leucine, Isoleucine and Valine

Attachment:

	U	C	A	G	
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	STOP	STOP	A
	Leu	Ser	STOP	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

54452.JPG

Question #: 10

A 56-year-old man comes to the physician because of a 2-day history of fever and a productive cough. His temperature is 39.8°C (100°F) and respirations are increased. An x-ray of the chest shows bilateral consolidation confirming pneumonia. Culture and sensitivity studies show the growth of erythromycin sensitive bacteria. Erythromycin is administered intravenously. Which of the following processes is most likely inhibited by this antimicrobial agent?

- A. Peptide bond formation during translation
- B. Initiation of translation
- C. Initiation of transcription
- D. DNA replication and cell division
- ✓E. The translocation step of translation
- F. Unwinding of double-stranded DNA

Rationale: Erythromycin inhibits the elongation step of translation specifically the translocation of the peptide chain from the A site to the P site.

Streptomycin inhibits the initiation of translation.

Chloramphenicol inhibits the peptidyl transferase activity on the bacterial (prokaryotic) ribosomes.

Eukaryotic peptidyl transferase activity is inhibited by cycloheximide.

Fluoroquinolones (Ciprofloxacin) inhibits bacterial DNA gyrase activity and inhibits replication.
Rifampin inhibits bacterial RNA polymerase and inhibits the initiation of transcription.
Alpha-amanitin inhibits eukaryotic RNA polymerase-II.
Topoisomerase inhibitors prevent unwinding of ds-DNA and hence inhibit replication.

Question #: 11

An 8-month-old boy is brought to the physician by his mother because he is unable to sit without support. Physical examination shows a loss of motor skills and a delay in the developmental milestones. Ophthalmoscopy shows a 'cherry-red' spot on the retina. Laboratory studies show no detectable beta-hexosaminidase A activity and a diagnosis of Tay Sachs disease is made. Targeted gene mutation analysis shows a frameshift mutation in the gene encoding beta-hexosaminidase A. Which of the following mutations is most likely in this patient?

- A. Insertion of three nucleotides in an exon
- B. Deletion of three nucleotides in an exon
- C. Substitution of C for an A in an exon
- ✓D. Insertion of four nucleotides in an exon
- E. Insertion of one nucleotide in an intron

Rationale: Insertion or deletion of nucleotides in multiples of three results in an in-frame mutation. Insertion of 4 nucleotides in an exon (not a multiple of three) results in a frame-shift mutation.

Question #: 12

A researcher studying carbohydrate metabolism isolates a bacterial strain which has a mutation in a gene encoding an enzyme required for fructose metabolism. There is no detectable enzyme activity in cell lysate from this bacterial strain. DNA sequencing of the gene encoding this protein shows that a UCA codon was mutated to a UAA codon. Which of the following best explains why this mutation led to a loss of enzyme activity in this bacterial strain?

- A. Ribosome is no longer able to initiate protein synthesis
- B. A frame shift was caused by the mutation
- C. An incorrect amino acid is placed in the enzyme active site
- ✓D. A shorter than normal protein is synthesized
- E. There is defective splicing of the introns
- F. The mRNA is shorter than normal

Rationale: The UCA codon which codes for Serine is replaced by UAA which is a termination codon and results in premature termination of translation of the enzyme. This is an example of a nonsense mutation.

Question #: 13

An investigator is studying the kinetic properties of an enzyme in the presence of a newly discovered drug that acts as an enzyme inhibitor. The Lineweaver-Burk plots for the uninhibited reaction (no inhibitor added) and differing concentrations of the specific inhibitor is shown. Which of the following statements best describes the data shown in line 1?

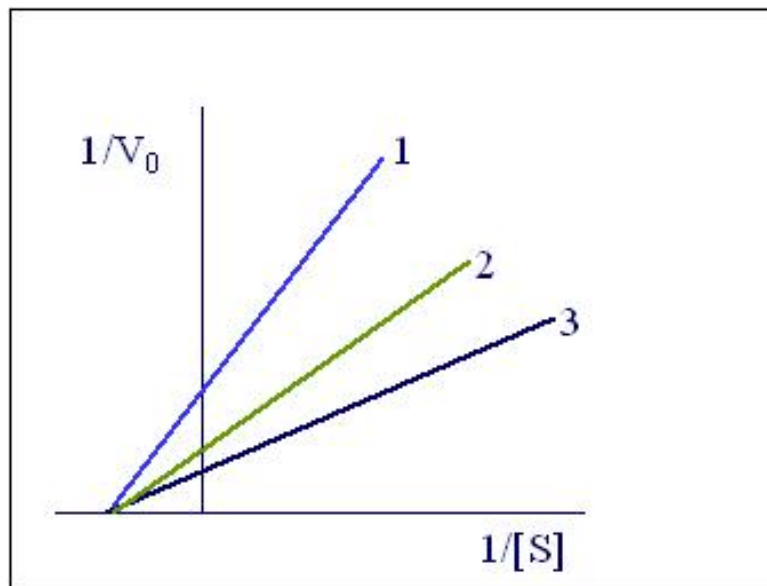
- A. No inhibitor added
- B. Highest concentration of a competitive inhibitor
- ✓C. Highest concentration of a noncompetitive inhibitor
- D. Lowest concentration of a competitive inhibitor
- E. Lowest concentration of a noncompetitive inhibitor

Rationale: Line 3 - Uninhibited reaction with K_m on X-axis and highest V_{max} ; the line with the least slope is the Uninhibited reaction (control);

Line 2 - V_{max} is lower and K_m is unchanged - Indicates lower concentration of a non-competitive inhibitor;

Line 1 - has a very low V_{max} and K_m is unchanged; represents highest concentration of a non-competitive inhibitor

Attachment:



4243.JPG

Question #: 14

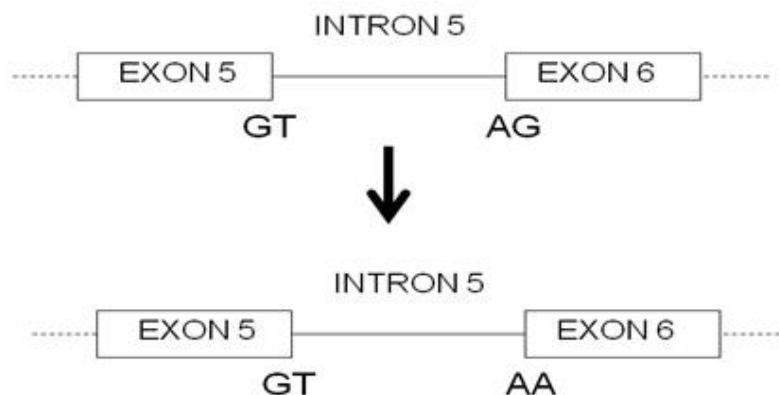
A 10-year-old boy is brought to the physician by his mother because of a 6-month history of progressive weakness of his arms and legs. Physical examination shows reduced muscle

strength in the arms and hips and atrophy of the deltoid and gluteus muscles. DNA is prepared from a muscle biopsy obtained from the patient and regions of dysferlin are amplified by PCR. A mutation in dysferlin is identified and a diagnosis of Limb-Girdle Muscular Dystrophy 2B is made. The dysferlin gene is found to have an AG → AA alteration at position 1686 at the splice acceptor site (3' splice site) on intron 5 as shown in the figure. Which of the following is the most likely effect of this mutation?

- A. Exon 5 is absent in the mature mRNA
- B. The mature mRNA contains extra exons
- C. There is no change in the mature mRNA
- D. The mRNA has a premature stop codon in exon 5
- ✓E. The protein has fewer amino acids

Rationale: As a result of the mutation in the splice acceptor site (3' splice site), the spliceosome looks for the next AG sequence (acceptor site) and as a result, a part of Exon 6 may be removed (spliced) resulting in the formation of a protein with fewer amino acids

Attachment:



26403.JPG

Question #: 15

A bacterial strain is being studied that is found to have reduced rates of DNA replication resulting in reduced rates of cell division. Activity of which of the following proteins is most likely impaired in this strain?

- A. Sigma factor
- ✓B. Single strand binding proteins (SSB)
- C. U1-snRNP
- D. Initiation factor-1 (IF-1)
- E. Peptidyl transferase

Rationale: SSB are the only proteins in the given choices that is involved in DNA replication. Sigma factor is required for transcription. U1 snRNP are required for splicing. IF-1 and peptidyl transferase are required for translation

Question #: 16

A researcher is analyzing the replicative DNA polymerase isolated from a novel bacterium. Studies show that the DNA polymerase from the organism lacks 3' to 5' exonuclease activity. Which of the following is most likely observed in this bacteria?

- A. Decreased mRNA synthesis
- ✓B. Increased mutation rate
- C. Ability to replicate DNA without a primer
- D. Inability to replicate DNA
- E. Inability to unwind the double helical DNA

Rationale: The 3'-5' exonuclease activity of the enzyme allows for the incorrect base pair to be excised. This activity is known as proofreading. Loss of proofreading leads to DNA mutations

Question #: 17

A 42-year-old woman comes to the physician because of a 4-day history of worsening cough and fever. Physical examination shows respiratory distress. Auscultation over the lungs shows diffuse crepitations and a diagnosis of lower respiratory tract infection is made. She is prescribed erythromycin for the infection. Which of the following processes is most likely inhibited by this drug?

- A. Binding of aminoacyl tRNA to A site of ribosome
- B. Unwinding of supercoiled DNA
- C. Initiation of translation
- ✓D. Translocation of peptidyl tRNA
- E. Peptidyl transferase activity of ribosome
- F. Transcription of mRNA

Rationale: Erythromycin inhibits protein synthesis (translation) in bacteria. It prevents translocation of the peptidyl t-RNA from the A-site to the P-site.

Binding of aminoacyl tRNA to A site of ribosome - Inhibited by tetracycline;

Unwinding of supercoiled DNA - inhibited by ciprofloxacin (quinolones) which inhibit DNA gyrase;

Initiation of translation - Inhibited by Streptomycin;

Translocation of peptidyl tRNA - Inhibited by erythromycin;

Peptidyl transferase activity of ribosome - Inhibited by chloramphenicol;

Question #: 18

During telophase at the end of mitosis, microtubules pull sister chromatids to opposite ends of the cell poles. Which best describes the portion of the chromosome that attaches to the microtubules?

- A. Euchromatin
- ✓B. Heterochromatin
- C. Telomeric
- D. The *Alu* sequence
- E. Pseudogene

Rationale: Microtubules attach to the centromeric region of chromosomes, and the centromere is composed of repetitive sequences which are compact areas of DNA and called heterochromatin. The centromere does not contain expressed genes. Euchromatin describes areas of the genome that have active gene expression. Telomeres are repetitive sequences that are found on the termini of chromosomes. The *Alu* sequence describes a DNA restriction endonuclease site that tends to be found within some transposable elements such as the SINE (short interspersed nuclear element). There are millions of copies of this repetitive sequence in the genome. Pseudogenes are DNA sequences that look like genes, but for some reason are not expressed.

Question #: 19

A scientist studying actively proliferating cultured cells uses a toxin to arrest the cells to block their exit from G2. Which of the following would best characterize these cells?

- A. Chromosomes are aligned at metaphase plate
- B. Nuclear membrane has dissolved
- ✓C. The DNA is duplicated
- D. Crossing over is occurring
- E. Cell is preparing to duplicate DNA

Rationale: In cells that are actively progressing through the cell cycle, preparation for the next cell division occurs in the G1 phase and DNA synthesis occurs in the S phase. Following the S phase, the cell enters G2 where the cell prepares for mitosis. If a cell culture is arrested so the cells cannot exit from G2, they will have duplicated DNA.

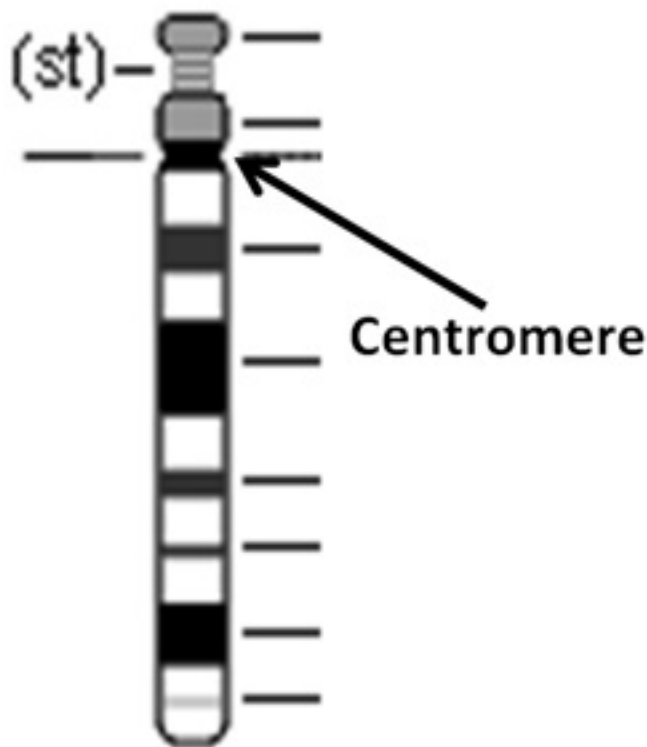
Question #: 20

A genetic test shows that the Down syndrome in a 4-year-old boy was caused by chromosome 21 undergoing a fusion to chromosome 14. Both of these chromosomes have a similar morphological structure as shown in the figure. Which of the following best describes the DNA that is found on the p arm of this type of chromosome?

- A. Methylated cytosine
- ✓B. Repetitive ribosomal RNA genes
- C. Intermingled RNA/DNA
- D. Continuous centromeric structure
- E. Continuous telomeric structure

Rationale: Chromosomes 13, 14, 15, 21, and 22 are called the acrocentric chromosomes. The acrocentric chromosomes contain a q arm (the larger arm) which are similar to a typical chromosome. The p arm of the acrocentric chromosome consists of an array of repetitive sequence that codes for ribosomal RNA. Living cells must create a lot of protein, to translate a lot of mRNA, the cell needs a lot of rRNA (to make ribosomes). The only way to get enough of this material is to have repetitive sequences of these genes. This is a typical strategy used by cells to get enough gene product. Another example is the histone genes (found elsewhere in the genome). The histone genes are also found as highly repetitive sequences, because actively dividing cells need a lot of histone mRNA, to make a lot of histone protein.

Attachment:



175206 ESoft Q fig ID.jpg

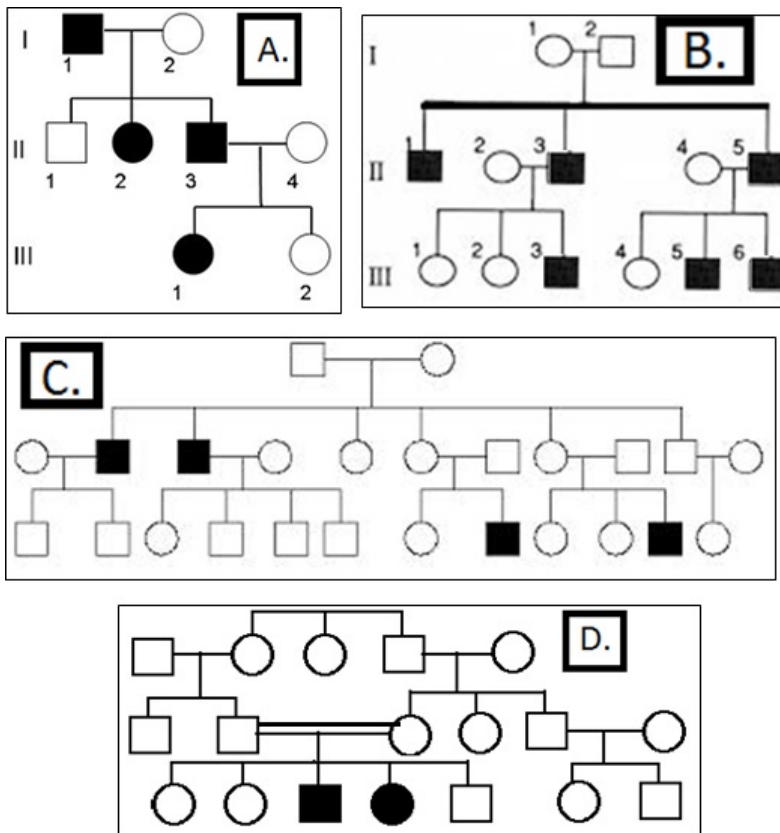
Question #: 21

An investigator studying Mendelian traits constructs several pedigrees to describe classic patterns of inheritance. Which of the following genograms would most likely describe an autosomal recessive disorder?

- A. The pedigree in panel A
- B. The pedigree in panel B
- C. The pedigree in panel C
- ✓D. The pedigree in panel D
- E. None of the pedigrees shown are suggestive of autosomal recessive

Rationale: The pedigree in panel D is most suggestive of autosomal recessive inheritance. Pedigree D shows presumably healthy parents who would be carriers, and children who have the disorder after inheriting a pathogenic variant from each parent. The consanguinity shown in panel D is also a clue, and the consanguinity of course increases the likelihood that there will be homozygosity by descent. Further things to think about with these four pedigrees: Pedigree A: What is the likelihood that any child of III-1 will inherit the disease associated allele? Which disorders are most likely associated with this pedigree? Assume that Pedigree B is not Y-linked; what indicates that Pedigree B cannot be X-linked? Pedigree C: Affected man (II-2) has another child, this time it is a girl. What is the risk that she will be a carrier? Which disorders would show this pattern? Who are the obligate carriers? Which females have a 50% risk of being a carrier? Why are there no male to male transmissions? Pedigree D: Besides being the parents of 5 children, what is the relationship between II-2 and II-3?

Attachment:



175209 ESoft Q fig ID.jpg

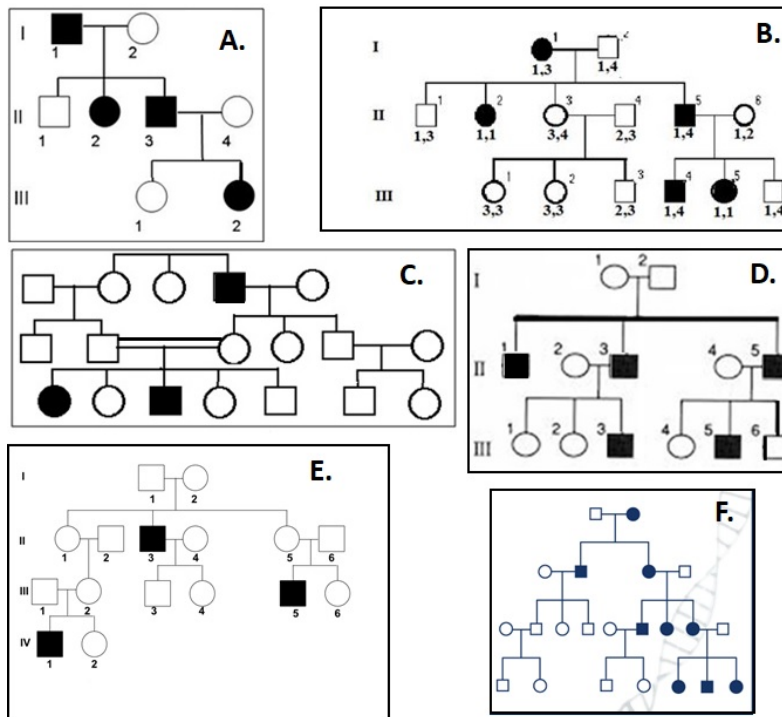
Question #: 22

A 22-year-old woman comes to the physician because she is concerned about a genetic condition that has affected some members of her family. Some of her relations have red-green color blindness and she wants to know if her children will be at risk of inheriting this trait. Which of the following pedigrees would most likely represent the family of this woman?

- A. Pedigree A
- B. Pedigree B
- C. Pedigree C
- D. Pedigree D
- ✓E. Pedigree E
- F. Pedigree F
- G. All of these pedigrees are equally likely to represent this family
- H. None of these pedigrees may represent this family

Rationale: Red-green color blindness is a classic X-linked trait. Only pedigree E shows transmission from carrier females to affected males.

Attachment:



91096 E Soft ID, for quiz.jpg

Question #: 23

A researcher finds that a protein truncation mutation in the lysine methyl transferase gene, *SETD6*, is responsible for a rare form of inherited colon cancer. The truncated *SETD6* protein is expressed in the same subcellular area, and it binds to its substrates as normal. However, the mutated protein blocks the ability of wild-type *SETD6* (expressed from the other allele) to function properly. This results in abnormal signaling and tumor development. Which of the following genetic terms best describes the mechanism of this inherited form of colon cancer?

- A. Autosomal recessive
- B. Haploinsufficiency
- ✓ C. Dominant negative
- D. Anticipation
- E. X-linkage
- F. Imprinting

Rationale: Explanation: from the vignette it should be clear that this is a dominant disorder since only one mutation needs to be inherited to cause the trait to manifest (colon cancer). It is not haploinsufficiency, because both alleles are expressed. It is not gain of function, because the encoded protein is not really doing anything new. The best answer is dominant negative, because the expressed protein from the gene with the mutation is blocking the ability of the normal gene from carrying out its normal function. Students often have confusion between “gain of function” and “dominant negative”. One way to think of this might be to consider that “dominant negative” is a special case of gain of function. In

dominant negative, the effect of the mutation is to interfere with the normal function of the gene. Certain types of osteogenesis imperfecta are explained by dominant negative as the presence of the mutant procollagen interferes with the formation of normal mature collagen.

Three examples of “gain of function” are listed: (1) – Huntington disease (encoded by HTT) – here the protein that has the expanded polyglutamine tract takes on a new function, leading to neurodegeneration; this is gain of function, because to cause neurodegeneration is not the normal function of the HTT gene. (2) Specific mutations in FGFR3 cause achondroplasia due to unregulated persistent signaling – causes dwarfism. (3) Autosomal recessive sickle cell disease occurs because the hemoglobin protein polymerizes (and distorts the red blood cell). Hemoglobin is supposed to exist as a discrete tetrameric protein. Normally, hemoglobin should not polymerize so the sickle cell allele (when homozygous) creates a “gain of function”. Interestingly, the Gly6Val mutation that causes SCD does not have any effect on the ability of Hb to bind to oxygen.

I don't advise it, but you may read more about SETD6 here: Martin-Morales, et al, 2017. SETD6 dominant negative mutation in familial colorectal cancer type X. Hum Mol Genet. 2017 Nov 15;26(22):4481-4493. doi: 10.1093/hmg/ddx336. <https://www.ncbi.nlm.nih.gov/pubmed/28973356>.

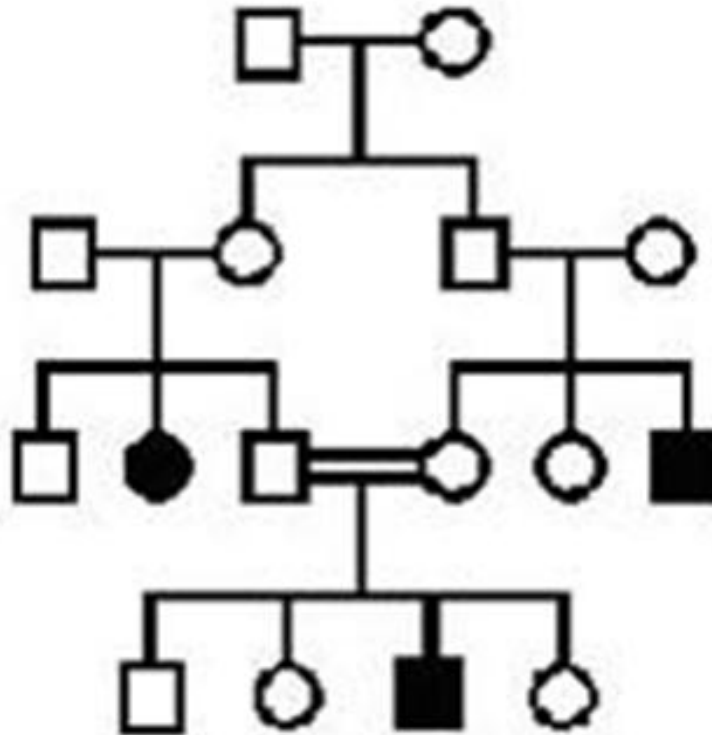
Question #: 24

A 42-year-old man (III-3) whose third child has phenylketonuria comes to a physician to learn about recurrent risk to his grandchildren as his first-born son now has a pregnant girlfriend. A pedigree of the family is shown (note that the pregnant girlfriend is not shown on this genogram). Which of the following would be the best estimate of the probability that IV-1 is a carrier of the mutation that leads to the inborn error of metabolism that was previously seen in this family?

- ✓A. 2/3
- B. $\frac{1}{2}$
- C. Virtually nil
- D. 100%
- E. $\frac{1}{4}$

Rationale: This is the 2/3 rule. The first person in the fourth generation is not affected, but his sibling is a known affected so both of the parents (III-3 and III-4) must be carriers. Since IV-1 is not affected, he may not be a/a, but he may be either A/A, a/A, or A/a. Two of these 3 options are carriers, so answer is 2/3.

Attachment:



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Question #: 25

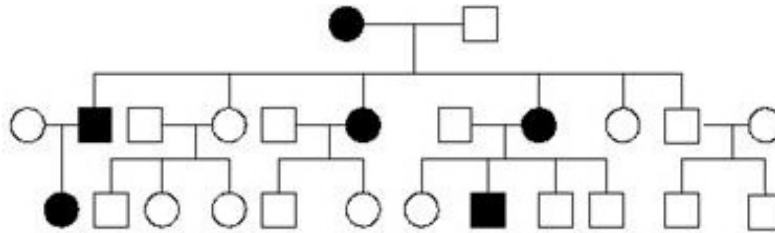
A 23-year-old man comes to the physician for a follow up consultation about a previously diagnosed genetic disorder. He has arachnodactyly, pectus excavatum, and he is taller than his siblings and cousins. A pedigree of this patient's family is shown. At this visit, the physician reviews the results from the latest echocardiogram because the patient is at risk for an aneurysm resulting from a dilated ascending aorta. Which of the following genes most likely has a pathogenic variant to cause the disorder in this patient?

- A. The *COL1A1* gene which encodes a pro-collagen chain
- B. The *FGF3R* gene which encodes fibroblast growth factor receptor 3
- C. The *LDLR* gene which encodes the LDL receptor
- ✓D. The *FBN1* gene which encodes fibrillin
- E. The *DMD* gene which encodes dystrophin
- F. The *HTT* gene which encodes huntingtin
- G. The *DMPK* gene which encodes dystrophin myotonic protein kinase
- H. The *HEXA* gene which encodes hexosaminidase A

Rationale: From the vignette it should be recognized that this patient has symptoms of Marfan syndrome which is caused by autosomal dominant mutations in the gene encoding fibrillin. Mutations in *COL1A1*

lead to osteogenesis imperfecta. Achondroplasia is caused by dominant “gain of function” mutations in FGFR3. Haploinsufficiency of LDLR causes familial hypercholesterolemia. Duchenne muscular dystrophy is caused by mutation of the X-linked DMD gene which encodes for the dystrophin protein. Huntington disease is caused by a triplet repeat expansion of a CAG codon in the first exon of the HTT gene. Myotonic dystrophy is also a triplet repeat expansion disorder that causes muscle weakness. Tay Sachs disease is an autosomal recessive inborn error of metabolism.

Attachment:



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Question #: 26

A 4-year-old boy is brought to the physician after worsening complaints of dizziness and headache. Laboratory tests show elevated insulin (hyperinsulinemia) and hypoglycemia. A genetic test shows a mutation in the gene encoding glucokinase. The effect of this mutation is predicted to mutationally alter the allosteric activator site of the enzyme. In which region of the gene would the disease-causing mutation most likely be found?

- A. Enhancer
- ✓B. Exon
- C. Intron
- D. Promoter
- E. Shine Dalgarno

Rationale: The allosteric site of an enzyme is part of the polypeptide polymer (the protein), so it must be encoded by the gene. Therefore, this sequence will be found on an exon of the mRNA, not an intron. Enhancers and promoters are DNA elements that bind to proteins to regulate gene expression. The Shine-Dalgarno sequence is found in bacterial genes and is thought to help ensure that ribosomes are correctly aligned with the RNA during translation.

Question #: 27

A 4-year-old girl is brought to the physician for follow-up examination by her parents because she shows phenotypic characteristics that are similar, but not identical, to Down Syndrome. Standard karyotype shows that in addition to trisomy 21, she has a large translocation where

approximately 50% of the q arm of chromosome 1 is attached to another chromosome. Which of the following genetic terms would best describe the appearance of the aberrant chromosome 1 following this translocation event?

- A. Acrocentric
- B. Telocentric
- C. Barr body
- ✓D. Submetacentric
- E. Metacentric

Rationale: Explanation: Chromosome 1 is defined as a metacentric chromosome because the centromere is found approximately in the middle of the chromosome. This is because the p and the q arm of chromosome 1 are of the approximate same size. If half (50%) of the chromosome 1 is removed, and attached to some other chromosome, then the resulting chromosomal structure would best be described as submetacentric.

Question #: 28

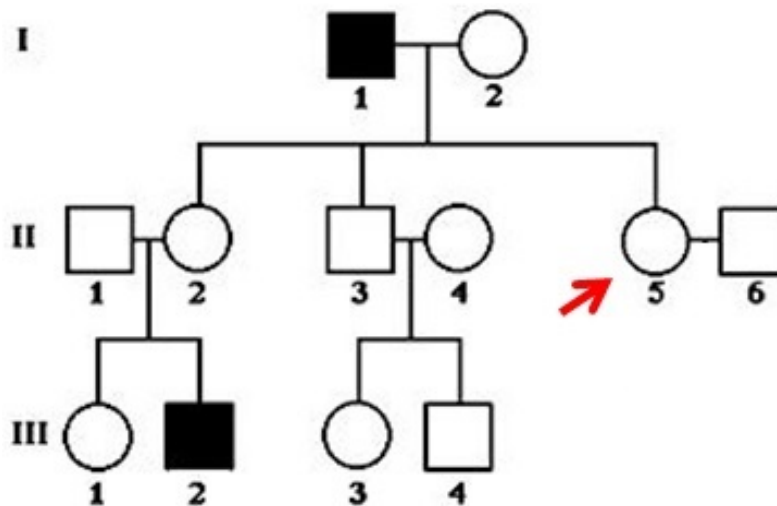
A 26-year-old woman (the consultand; II-5) comes to the physician because there is a history of a bleeding disorder due to deficiency of factor VIII in her family. She is now thinking about having a family of her own. A 3-generation pedigree of the immediate family is shown. Which of the following best explains why are there no affected family members in the second generation?

- A. Not enough female offspring in generation II (the sample size is too small)
- ✓B. Normal X-linked inheritance
- C. Effect of heteroplasmy
- D. Phenotype exhibits variable expressivity
- E. The disorder exhibits incomplete penetrance
- F. The disorder exhibits allelic heterogeneity

Rationale: The pedigree is suggestive of X-linked inheritance. Hemophilia A and B are both X-linked. All females of affected males must be carriers (and are likely to not be affected). All males of affected males must be unaffected. Therefore, this is normal X-linked inheritance.

Even with large sample sizes, it is unlikely to find a female who manifests an X-linked recessive disorder. If a female does show symptoms of an X-linked disorder, she is either a manifesting heterozygote, or she has inherited two loss of function mutations in the gene. Heteroplasmy describes variation in severity of mitochondrial disorders.

Attachment:



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Question #: 29

A married couple comes to the physician to learn about the recurrence risk of cystic fibrosis (CF) to their future children. They are both healthy and currently have 1 child with CF. Analysis of the *CFTR* alleles of the father shows that he is heterozygous for the Phe508del variant (F508). Analysis of the mother shows that she is heterozygous for a missense variant leading to an Arg117His substitution. The child inherited the disease associated alleles from both parents. Which of the following genetic terms best describes this example?

- A. Incomplete penetrance
- ✓B. Allelic heterogeneity
- C. Variable expression
- D. Heteroplasmy
- E. Locus heterogeneity
- F. Uniparental disomy

Rationale: CF is an autosomal recessive (AR) disorder caused by inheritance of loss of function mutations in the *CFTR* gene from both parents. There are many mutations that might cause loss of function; hence the *CFTR* gene exhibits allelic heterogeneity. If the two mutations are different (but both cause loss of function), the mutational profile in the affected person is described as being compound heterozygote.

Question #: 30

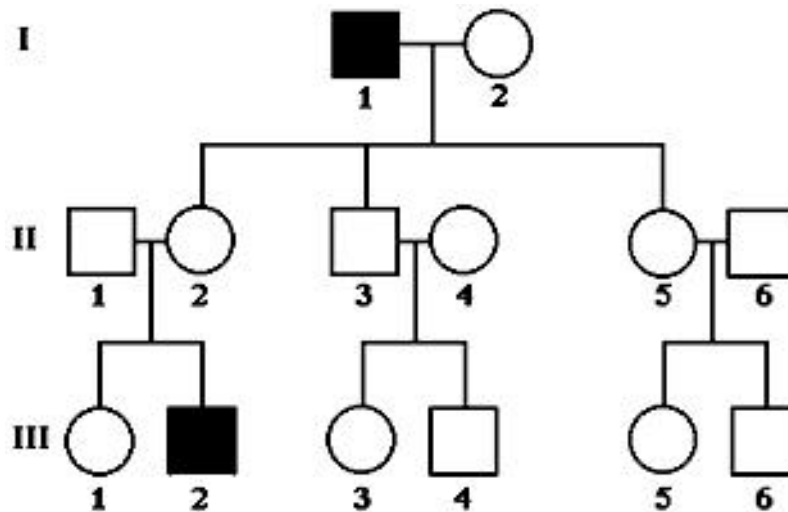
A 35-year-old woman comes to the physician for genetic counseling. She is a member of a family that has some individuals affected with Becker muscular dystrophy. Family history is

recorded, and a pedigree of the family is shown. Which of the following individuals are most likely the obligate carriers of the pathogenic variant in this family?

- A. I-2, II-2
- B. II-1, III-1
- ✓C. II-2, II-5
- D. III-5, II-4
- E. III-1, III-5
- F. II-2, II-3, II-5

Rationale: First you must recognize that this is an X-linked trait. Next you must see that the affected grandfather (I-1) must give his X chromosome to his daughters (they must be carriers). So II-2 and II-5 must have inherited the X-chromosome with the variant causative of the trait from their father, and they must be (obligate) carriers. Note that II-3 may not be a carrier because he only has one X chromosome, and he had to have inherited it from his mother (his father gave him the Y chromosome, and this is why he is male). You should also be able to make the following conclusion: III-1 and III-5 each have a 50% chance of being a carrier because we know that their mothers are carriers and they could have inherited either X chromosome from their respective mothers.

Attachment:



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